

Prick-free Blood Tester from India

Many of us are terrified of blood tests, where skin is pricked with a needle to get the blood for testing. A solution to this problem has come in the form of TouchB—a prick-free blood testing device



The inside of TouchB

ASHWIN GOPINATH

Say hello to TouchB—a prick-free blood testing device brought to you by Biosense Technologies. TouchB can measure your blood haemoglobin, oxygen saturation, temperature and pulse rate all without pricking for blood. A team of three—engineer Myshkin Ingawale and doctors Abhishek Sen and Yogesh Patil—was behind making this product from the ground up.

What led the team to come up with this idea? Myshkin narrates, “In 2009, I had gone to visit my friend Abhishek in Parol—a two-hour drive from Mumbai. He was interning at the

place and when I reached Parol, I had to wait for him as he was busy helping deliver a baby. Afterwards, he came out ashen-faced and informed me that both the mother and baby had died of something called post-partum haemorrhage, which is basically uncontrollable bleeding. What was shocking was that it was caused by undiagnosed anaemia.”

The news that someone could die of anaemia—a completely diagnosable and curable disease—shocked Myshkin and made him look for a solution. He asked his mother, a paediatrician, and found that testing for anaemia involves sending a blood sample through a Coulter counter. But

the nearest such machine was in Mumbai and it wasn't a viable solution to keep using the machine to have every person tested for anaemia. He contacted his two friends and started searching for practical designs for the device.

Dr Sen says, “In India, public healthcare runs on the ASHA worker and not on the doctor. So when our team convened to design a solution for this problem, we knew that whatever we created had to be prickless (to avoid medical waste), simple enough for the ASHA worker to operate and small enough for her to carry in her kit.”

TouchB features

TouchB is a handheld battery-operated device.

Total blood haemoglobin is used for the diagnosis of anaemia. (The WHO qualifies any pregnant woman with Hb level of less than 11 grams per decilitre of blood as anaemic.)

TouchB works on the principle of ‘photoplethysmography.’ Basically, it works by radiating light of three different wavelengths onto the finger, and through the tissue of the patient's finger. Once this is done, based on the amount of light transmitted, absorbed and scattered, one can figure out the amount of haemoglobin in blood. Haemoglobin has a characteristic absorbance. For estimation of oxygen saturation, the technique is similar to the one used in pulse oximeters.